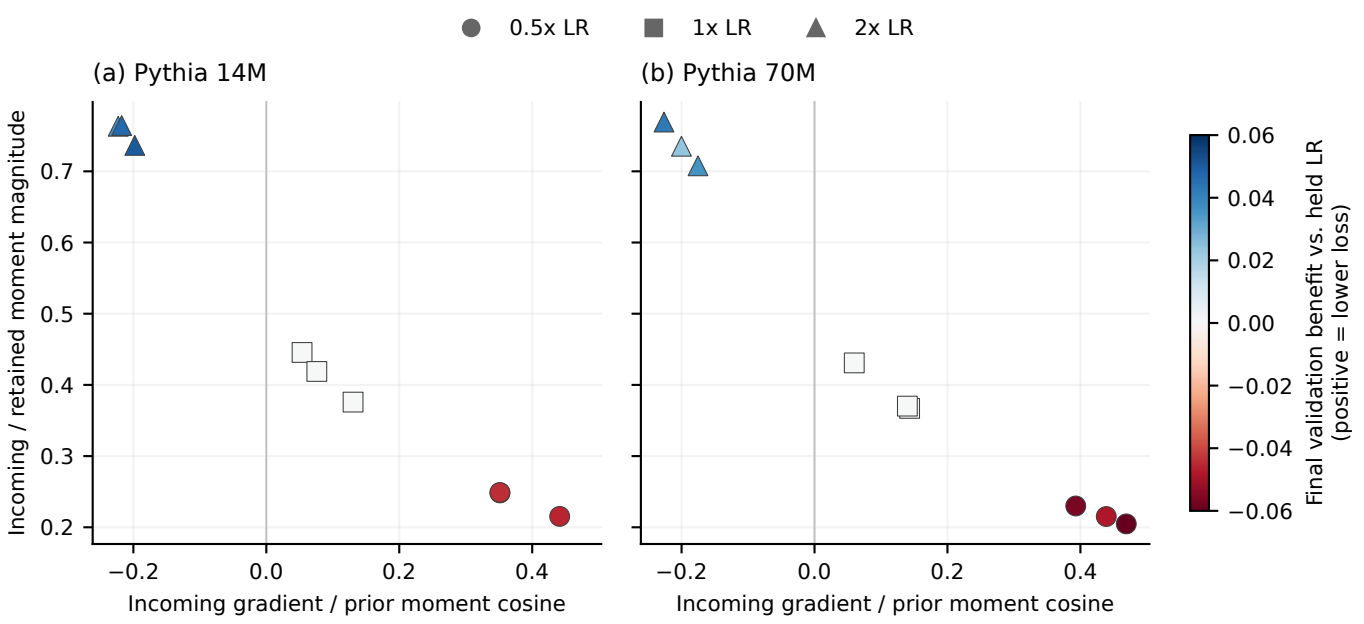

0.5x LR
1x LR
2x LR
(a) Pythia 14M
(b) Pythia 70M
Incoming / retained moment magnitude
0.7
0.6
0.5
0.4
0.3
0.2
−0.2
0.0
0.2
0.4
Incoming gradient / prior moment cosine
Incoming gradient / prior moment cosine
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
Final validation benefit vs. held LR
(positive = lower loss)